

## Ideation Phase

### Define the Problem Statements

|               |                                                                 |
|---------------|-----------------------------------------------------------------|
| Date          | 09 March 2026                                                   |
| Team ID       |                                                                 |
| Project Name  | Electric Vehicle Data Analytics and Visualization using Tableau |
| Maximum Marks | 2 Marks                                                         |

| Problem Statement (PS)                          | I am (Customer)                                                                               | I'm trying to                                                                                                                                                                  | But                                                                                                               | Because                                                                                                                   | Which makes me feel                                                                                                                    |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| PS-1 (Charging Pattern Analysis – City Planner) | I am a <b>city energy planner</b> responsible for managing electricity demand in urban areas. | I'm trying to <b>understand when and where electric vehicle users charge their vehicles</b> so that I can plan the electricity supply and charging infrastructure efficiently. | But it is <b>difficult to identify charging patterns</b> because the data is scattered and not easy to interpret. | Because <b>EV charging demand changes</b> depending on time, location, and user behaviour.                                | Which makes me feel <b>uncertain about whether the existing power infrastructure will be able to handle future EV charging demand.</b> |
| PS-2 (Battery Performance – Fleet Manager)      | I am a <b>fleet manager</b> managing multiple electric taxis in the city.                     | I'm trying to <b>monitor battery performance</b> and understand how it affects the actual driving range of my vehicles.                                                        | But it is <b>challenging to track battery health</b> and predict performance using raw vehicle data alone.        | Because <b>battery efficiency changes</b> over time depending on charging cycles, driving conditions, and usage patterns. | Which makes me feel <b>concerned about unexpected vehicle downtime</b> and increased maintenance costs.                                |